Homework 2 Isabella Pensabene irp28

1) <u>known</u>: $R = 0.5m$ $H = 1m$ $T_w = 50°C$ $T_s = 50°C$ $T_\infty = 20°C$ $h = 15\ W/m^2K$

<u>assumptions</u>: steady state, thin wall $\rightarrow T_s = T_w$, no radiation, bottom of heater well insulated

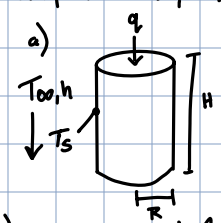
b) 1D heat conduction through foam,

<u>find</u>: a) electrical heating power input $q = ?$ at steady state

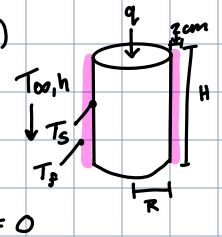
b) w/ $L = 2cm$ foam w/ $k = 0.1\ W/mK$ find $q = ?$

c) compare a) & b)

<u>schematic</u>: a)



b)



<u>analysis</u>: a) $q_{total} = q - hA(T_\infty - T_s) = 0$

$A = 2\pi RH + \pi R^2 = 2\pi(0.5m) + \pi(0.5m)^2 = 3.927\ m^2$

$q = (15\ W/m^2K)(3.927\ m^2)(20°C - 50°C) \rightarrow \boxed{q = 1767.146\ W}$

b) $E_{in} = E_{out}$ for each surface:

$q = kA\frac{T_s - T_f}{L} = hA(T_f - T_\infty)$

$(0.1\ W/m^2K)\frac{323K - T_f}{0.02m} = (15\ W/m^2K)(T_f - 293K)$

$32.3 - 0.1T_f = 0.3T_f - 87.9 \rightarrow \boxed{T_f = 300.5K}$

$q = kA\frac{T_s - T_f}{L} = (0.1\ W/m^2K)(3.927\ m^2)\frac{278K - 300.5K}{0.02m}$

$\hookrightarrow \boxed{q = 441.7875\ W}$

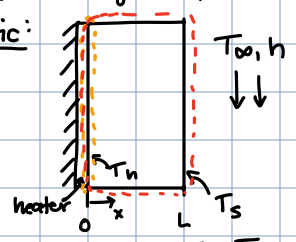
c) There is less required energy input when there is a layer of insulating foam than when there is no insulating layer.

2) **known**: $L = 10\,cm = 0.1\,m$ $k = 1\,W/(m\cdot K)$ $T_\infty = 20°C$ $h = 100\,W/(m^2K)$

a) $q'' = 1000\,W/m^2$ b) $T_s \le 200°C$

assume: 1D system, heater infinitely thin, no thermal radiation, steady state

schematic:



find: a) $T_h = ?$ & $T_s = ?$

b) $q''_{max} = ?$

c) 3 strategies to have more q''

analysis: a) $q'' = -k\dfrac{T_s - T_h}{L} = h(T_s - T_\infty)$

$\hookrightarrow T_s = \dfrac{q''}{h} + T_\infty = \dfrac{1000\,W/m^2K}{100\,W/m^2K} + 293.15K \rightarrow \boxed{T_s = 303.15K}$

$\hookrightarrow T_h = \dfrac{q''L}{k} + T_s = \dfrac{(1000\,W/m^2K)(0.1m)}{1\,(W/m^2K)} + 303.15K \rightarrow \boxed{T_h = 403.15K}$

b) $q'' = h(T_s - T_\infty) \rightarrow T_s = \dfrac{q''}{h} + T_\infty$

$\hookrightarrow T_s = T_h - \dfrac{q''L}{k}$

$T_h + \dfrac{q''L}{k} = \dfrac{q''}{h} + T_\infty$

$T_h = q''\left(\dfrac{1}{h} + \dfrac{L}{k}\right) + T_\infty \le T_{s\,max}$

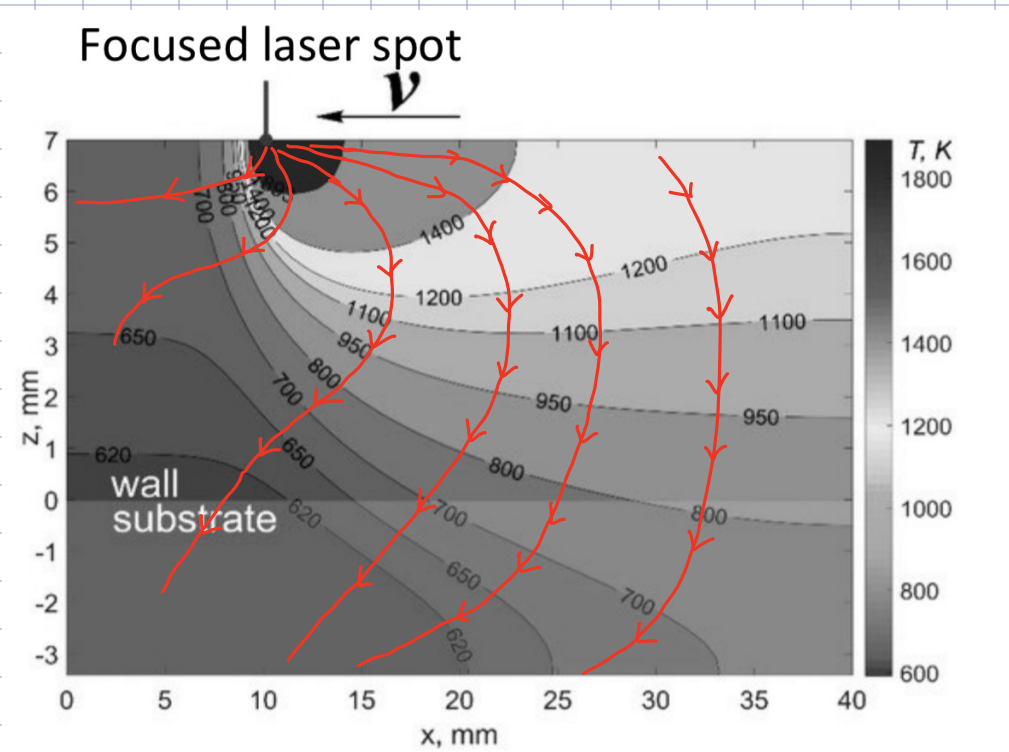
$\rightarrow q'' \le \dfrac{T_{s,max} - T_\infty}{\left(\frac{1}{h} + \frac{L}{k}\right)} = \dfrac{473.15K - 293.15K}{\left(\frac{1}{100} + \frac{0.02}{1}\right)} \rightarrow \boxed{q''_{max} = 1636.36\,W}$

c) 1) decrease thickness of insulation

2) decrease temperature of coolant

3) change coolant to increase h

3) a)



b) the left side has larger heat flux because the temperature is changing more in a shorter distance ($\frac{dT}{dx}$ is greater), so the heat flux is great